

Bridge 10-152 Course Names and Descriptions v1

Purpose

This artifact provides a staff-reference draft for the finalized bridge-course names and updated descriptions for the 10-152-nnn sequence only.

Design constraints applied:

- bridge course names are kept at or under 42 characters
- descriptions are written to remain under 700 characters
- names avoid unnecessary use of AI
- descriptions are written with enough flexibility to allow content adaptation as technology changes
- prerequisite course relationships are included for staff reference
- semester grouping follows the current bridge design direction, including moved and repurposed courses from the latest bridge summary

Semester 1 - Foundations and Controlled Entry

Total 10-152 semester credits: 8

Course No.	Bridge Name	Credits	Lec/Lab	Pre-requisite Course(s)	Draft Description
10-152-117	Python Programming Foundations	2	18/36	n/a	Learners explore Python programming by designing and implementing solutions to real-world problems. Students develop foundational programming skills using variables, data types, conditionals, loops, functions, and data structures. The course emphasizes problem-solving, debugging, and code comprehension while introducing the use of AI-assisted tools to support development. Learners practice building, evaluating, and refining programs both independently and with guided assistance. By the end of the course, students are able to create, understand, and explain basic Python programs and apply structured thinking to solve problems.
10-152-118	Web Development Foundations	3	36/36	n/a	Learners develop foundational web development skills using HTML, CSS, and JavaScript to create structured, styled, and interactive web pages. The course emphasizes building multi-page websites, applying design principles for usability and readability, and implementing basic interactivity through script-based behaviors. Learners practice debugging and refining web solutions using progressive techniques and are introduced to AI-assisted tools to support development. By the end of the course, students are able to design, build, and explain functional web solutions for varied contexts and needs.
10-152-119	Algorithmic Problem Solving	3	36/36	10-152-117	Learners are introduced to algorithms as structured approaches to solving problems. The course focuses on analyzing problems, selecting appropriate strategies, and implementing solutions using common data structures. Students explore how different approaches affect performance through basic analysis techniques and comparison of solutions. Emphasis is placed on reasoning, debugging, and refining solutions both manually and with AI-assisted tools. By the end of the course, learners are able to evaluate, adapt, and explain algorithmic approaches used to solve problems effectively.

Semester 2 - Data, Stack Depth, and Modern Development

Total 10-152 semester credits: 12

Course No.	Bridge Name	Credits	Lec/Lab	Pre-requisite Course(s)	Draft Description
10-152-120	Database Query and Design	3	36/36	n/a	Learners develop database skills by working with data models, structured queries, and relational design concepts. The course emphasizes organizing information, defining relationships, and constructing valid queries to create, retrieve, and manage data. Students practice using database tools to build tables, enforce keys, join related data, and generate useful outputs. The course is designed to support evolving platforms and workflows while strengthening the learner's ability to reason about how data is structured, accessed, and maintained in modern software systems.
10-152-121	Advanced Python Systems	3	36/36	10-152-117, 10-152-118	Learners deepen their Python skills by designing more capable, maintainable, and efficient software solutions. The course emphasizes structured program design, reusable components, data handling, testing, profiling, and the use of widely adopted libraries and development tools, including structured data handling and ORM-style application workflows. Students work with larger workflows involving files, datasets, automation, and analysis while practicing debugging, documentation, and code quality techniques. By the end of the course, learners are able to build and explain more advanced Python-based systems and adapt their approach to a range of technical problems and environments.
10-152-122	C# Application Development	3	36/36	10-152-117	Learners develop application-development skills using C# and related web application concepts to design and implement interactive software solutions. The course emphasizes core programming structures, object-oriented design, event-driven behavior, collections, file handling, exception management, and data access, while also introducing web-oriented application patterns. Students practice debugging, extending, and refining applications in ways that can adapt to changing tools and frameworks. By the end of the course, learners are able to build, explain, and improve applications that integrate both software logic and user-facing functionality.
10-152-123	Modern Data Modeling for Systems	3	36/36	10-152-120	Learners examine modern approaches to organizing, modeling, and managing data for software systems. The course emphasizes selecting appropriate data structures and storage patterns, working across relational and non-relational models, and understanding how data design affects reporting, integration, security, and scalability. Students compare alternative approaches, transform and interpret data, and practice reasoning about how information should be represented for different technical and business needs. By the end of the course, learners are able to explain and apply data-modeling choices in contemporary software environments.

Semester 3 - AI-Injected Architecture and Applied Context

Total 10-152 semester credits: 13

Course No.	Bridge Name	Credits	Lec/Lab	Pre-requisite Course(s)	Draft Description
10-152-124	Containerized App Deployment	2	18/36	n/a	Learners develop practical deployment skills by creating, configuring, and managing containerized application environments. The course emphasizes packaging software, organizing dependencies, and supporting consistent execution across development and delivery contexts. Students practice building images, running services, and refining container-based workflows through hands-on exercises. By the end of the course, learners are able to explain how containerization supports modern software delivery and apply those concepts to evolving tools, platforms, and deployment scenarios.
10-152-126	Cloud Platforms for Developers	3	36/36	10-152-124	Learners explore cloud development practices by working with platform services, scalable application patterns, and managed infrastructure concepts. The course emphasizes selecting and using cloud resources, understanding service integration, and supporting reliable application delivery in hosted environments. Students practice working with modern developer workflows, managed tools, and data services while examining tradeoffs related to scale, availability, and maintainability. By the end of the course, learners are able to explain and apply cloud-development patterns that can adapt to changing providers and platform ecosystems.
10-152-127	Systems Analysis and Design	3	36/36	n/a	Learners apply systems analysis and design methods to investigate problems, define objectives, evaluate constraints, and develop solution plans. The course emphasizes structured requirements work, workflow analysis, data and process definition, and responsible use of modern automation and AI-supported capabilities within larger systems. Students practice comparing solution paths, documenting decisions, and aligning technical choices with organizational needs. By the end of the course, learners are able to frame problems clearly, justify design decisions, and explain how systems can evolve responsibly as technologies and business requirements change.
10-152-128	C++ Systems Performance Foundations	2	18/36	10-152-122	Learners build systems-level programming foundations using C++ to explore how software behavior is shaped by data representation, control flow, and performance-aware design. The course emphasizes structured logic, input and output, data types, branching, and problem solving while introducing lower-level reasoning that supports broader understanding of runtime behavior. Students practice writing, debugging, and refining programs in ways that strengthen computational thinking across languages and platforms. By the end of the course, learners are able to explain and apply foundational systems-programming concepts in practical development contexts.
10-152-131	Machine Learning Foundations	3	36/36	10-152-121	Learners explore foundational machine learning concepts by working with common model types, sample datasets, and practical evaluation methods. The course emphasizes understanding how models are selected, trained, tested, and interpreted in support of software and decision-making tasks. Students examine expectations, compare outcomes, and evaluate effectiveness using real-world examples while also considering the role of AI-assisted tools in model-oriented workflows. By the end of the course, learners are able to explain basic machine learning processes, apply common techniques, and assess results in context.

Semester 4 - Delivery, Integration, and Demonstration

Current finalized 10-152 semester credits in this draft: 13

Course No.	Bridge Name	Credits	Lec/Lab	Pre-requisite Course(s)	Draft Description
10-152-129	Agile Delivery and DevOps	3	36/36	10-152-122, 10-152-127	Learners develop team-based delivery skills by planning, building, and refining software in a collaborative project environment. The course emphasizes SDLC awareness, version control systems, review workflow, light project management and team coordination practices, and shared responsibility for software quality. Students use current collaboration and delivery tools to simulate professional development practices while adapting to changing requirements and technical constraints. By the end of the course, learners are able to contribute to structured team workflows and explain how delivery practices support reliable software outcomes.
10-152-130	Software Career Experience	2	0/0*	10-152-127	Learners gain applied professional experience through mentored participation in workplace, field-based, simulated, service, capstone, or industry-related settings. The course emphasizes professional communication, problem solving, accountability, reflection, and the ability to connect prior technical learning to practical environments. Students document work, evaluate outcomes, and explain how their experience supports continued growth and employability. By the end of the course, learners are able to describe and demonstrate how their skills transfer into real-world software and technology contexts.
10-152-132	Mobile Application Development	2	18/36	n/a	Learners explore mobile application development by designing and implementing software for device-based and cross-platform environments. The course emphasizes user interaction, platform-aware design, structured development practices, and integration with device capabilities and external services. Students practice building, testing, and refining mobile solutions while considering usability, maintainability, and changing platform expectations. By the end of the course, learners are able to create and explain mobile application solutions that respond to real-world user and technical requirements.
10-152-133	Deep Learning Frameworks and Applications	3	18/72	10-152-131	Learners explore deep learning frameworks and applications by building and evaluating model-based solutions for tasks such as classification, prediction, and pattern recognition. The course emphasizes tensors, model architecture, training workflows, evaluation methods, and practical use of widely adopted development frameworks. Students compare implementation approaches, experiment with model behavior, and refine solutions based on evidence and performance goals. By the end of the course, learners are able to explain and apply deep learning workflows in ways that can adapt as tools, frameworks, and application areas evolve.
10-152-xxx or 10-156-xxx	Technical Intelligence Studio	3	18/72	10-152-121, 10-152-127, 10-152-131	Learners investigate an emerging technology area, evaluate its relevance, and complete a bounded implementation or demonstrable use case under instructor guidance. The course combines technology monitoring, structured analysis, experimentation, communication, and reflection into a single applied studio experience. Students present findings, explain implications for software practice, and develop a personalized learning plan that supports continued growth in a changing technical landscape. By the end of the course, learners are able to analyze, implement, demonstrate, and justify technology choices in a professional context.

Working Notes

- Semester 3 reflects the current bridge decision to move 10-152-131 earlier so machine learning foundations occur before deep-learning work.
- Semester 4 reflects the current bridge decision to repurpose 10-152-133 as **Deep Learning Frameworks and Applications**.
- 10-152-125 is intentionally absent because its content is being folded into 10-152-122 and the standalone bridge course is being dropped.
- The final Semester 4 studio course combines the earlier emerging-technology and studio concepts into a single bridge-usable course because the separate one-credit feeder course is not part of the bridge program.
- The full proposed title “Technical Intelligence Studio for AI Engineers” exceeds the 42 character bridge-name limit, so the bridge-safe short form **Technical Intelligence Studio** is used here.
- * 10-152-130 has 144 occupational hours in the current bridge. The draft description is written to allow for a range of applied experience types, including but not limited to workplace experience, field-based projects, simulations, service learning, capstone projects, and industry-related engagements. The final bridge design will determine the specific experience types included in the bridge offering.